

Variables are your friends, and friends of each other...?

Sociology 310 Discussions, Week 3

Agenda

- Assignment #1 Feedback
- Variables - mediators, moderators, confounders
 - to differentiate
 - to identify
- Individual Work - Assignment #2a
- Check Out

Note

- I noticed that a few students may need more information on what we covered last time - you're welcome to visit my OH!

on your Assignment #1...

Overall Comment

You're doing a nice job!

- Introductions provide clear contexts, such that the projects sound motivating
- RQs all sounds interesting and promising
- Conceptualizations contain helpful information to define your terms

Overall Comment

There're also improvements needed...

- proper **format** matters (e.g., references, spaces before/after, etc.)
- think about **logic flows**
 - things you mentioned above has a meaning for words below
 - don't make things "suddenly appear" but prepare your readers for them
- **cite** enough and necessary prior studies

Overall Comment

Visit my OH!

Variables

Variable Differentiation

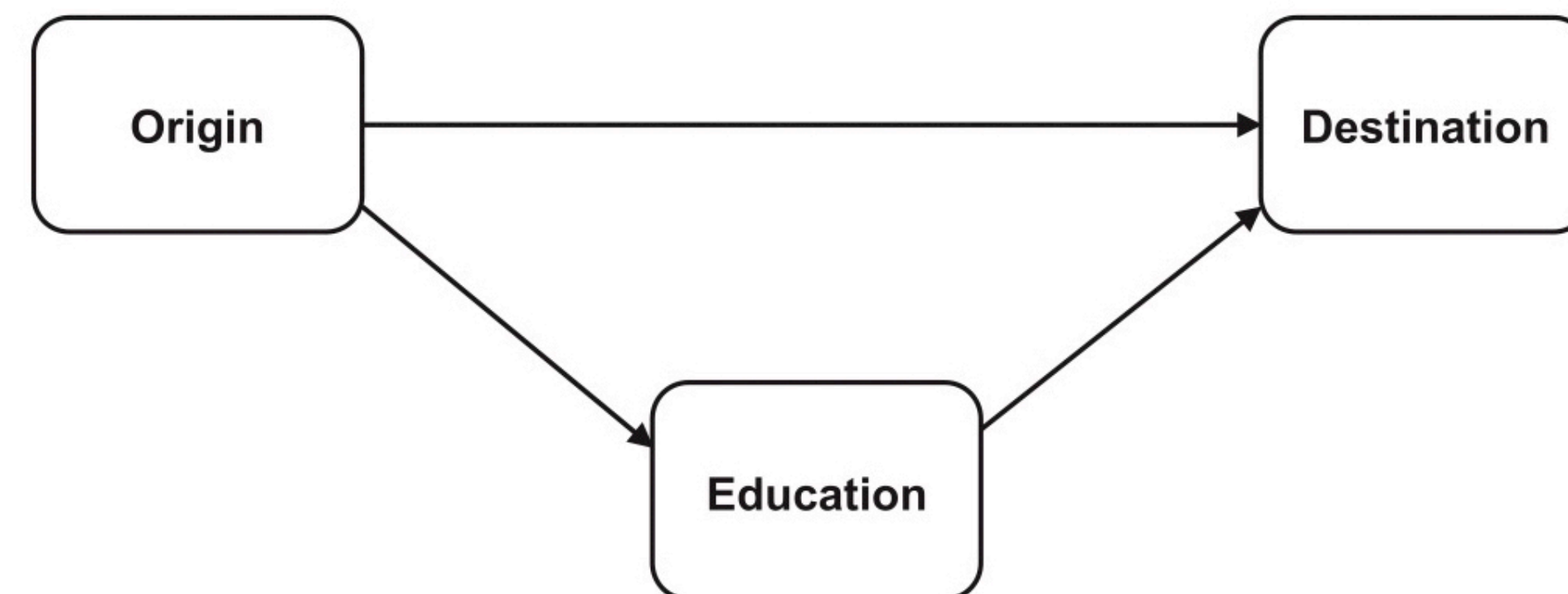
IV & DV

- Independent Variable - cause; explanatory variable; usually X
- Dependent Variable - outcome; response variable; usually Y
- $X \rightarrow Y$ denotes a causation relationship

Variable Differentiation

Mediators

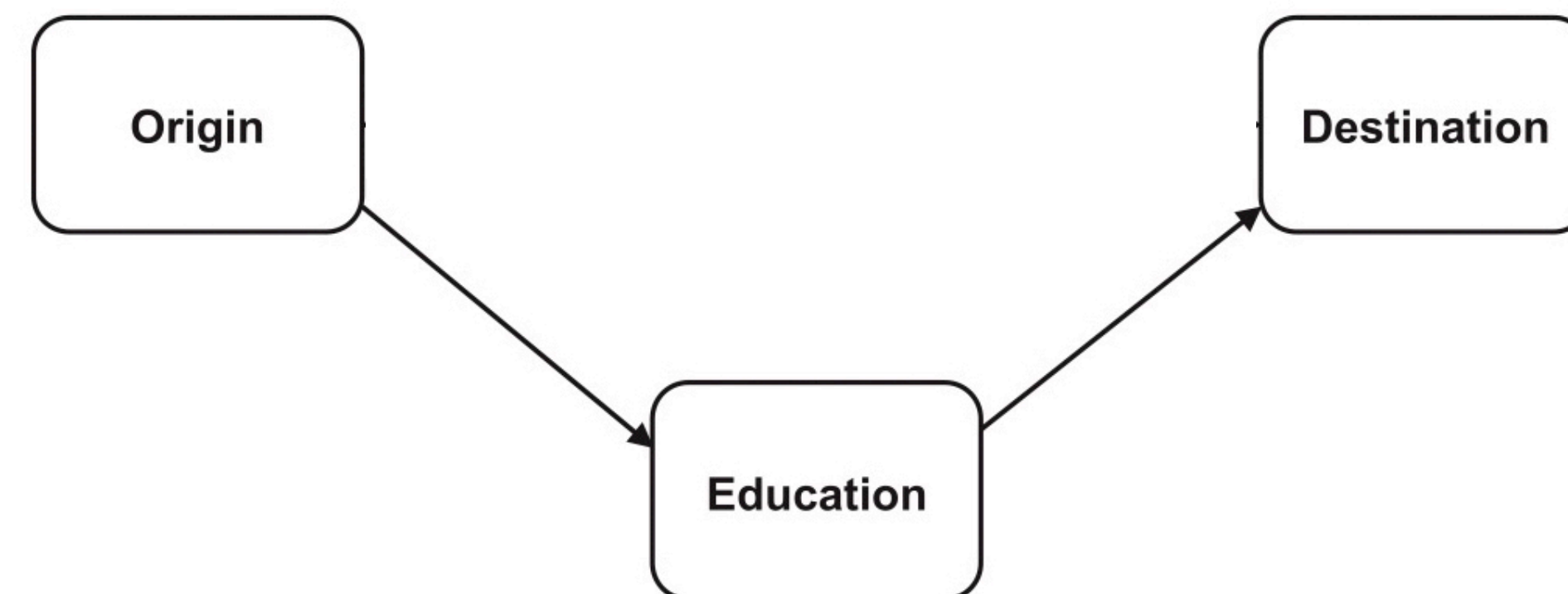
- asks “how?” or “by what means?” for $X \rightarrow Y$ causation
so something like M in $X \rightarrow M \rightarrow Y$ causation
- essentially intermediate variables explaining the mechanism(s) behind IV influencing DV
- similar to a bridge or a chain



Variable Differentiation

Mediators

- asks “how?” or “by what means?” for $X \rightarrow Y$ causation
so something like M in $X \rightarrow M \rightarrow Y$ causation
- essentially intermediate variables explaining the mechanism(s) behind IV influencing DV
- similar to a bridge or a chain
- full mediation:



Variable Differentiation

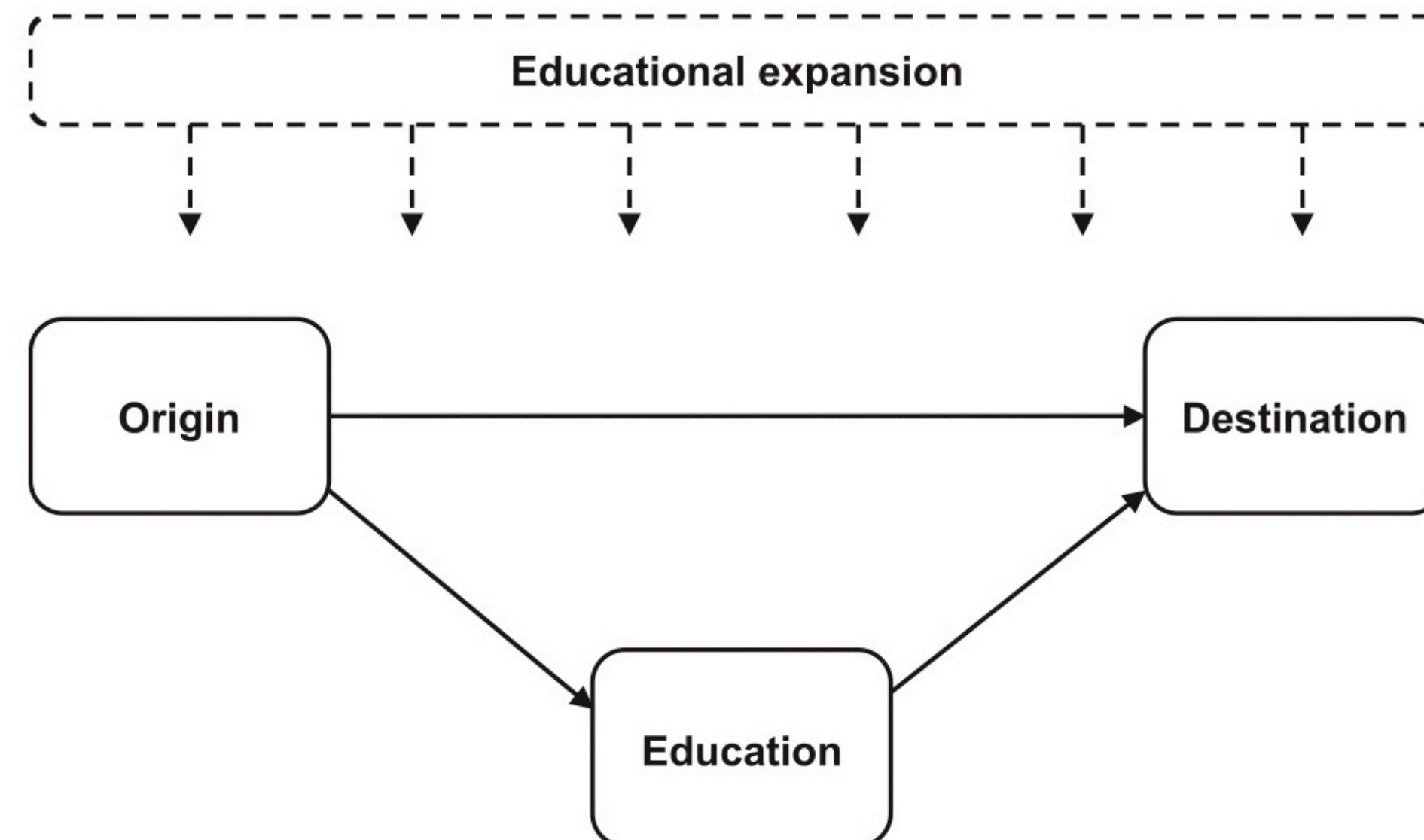
Moderators

- asks “when?” “for whom?” or “under what conditions?” for $X \rightarrow Y$ causation
- essentially variables altering the relationship between IV and DV
 - the existence of the influence(s)
e.g., something’s function may appear in one context but not in another
 - the degree/strength/size of the influence(s)
e.g., $V(\text{O}_2)$ from $V(\text{H}_2\text{O}_2)$ in the reaction $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2 \uparrow$ within a specific amount of time, with and without MnO_2 as catalyst
- similar to a switch

Variable Differentiation

Moderators

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- essentially variables altering the relationship between IV and DV



Variable Differentiation

Confounders

- asks “what else?” for $X \rightarrow Y$ *association*
- essentially variables correlated with IV, causing DV, and *not* a mediator
- similar to a puppet master

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Confounders

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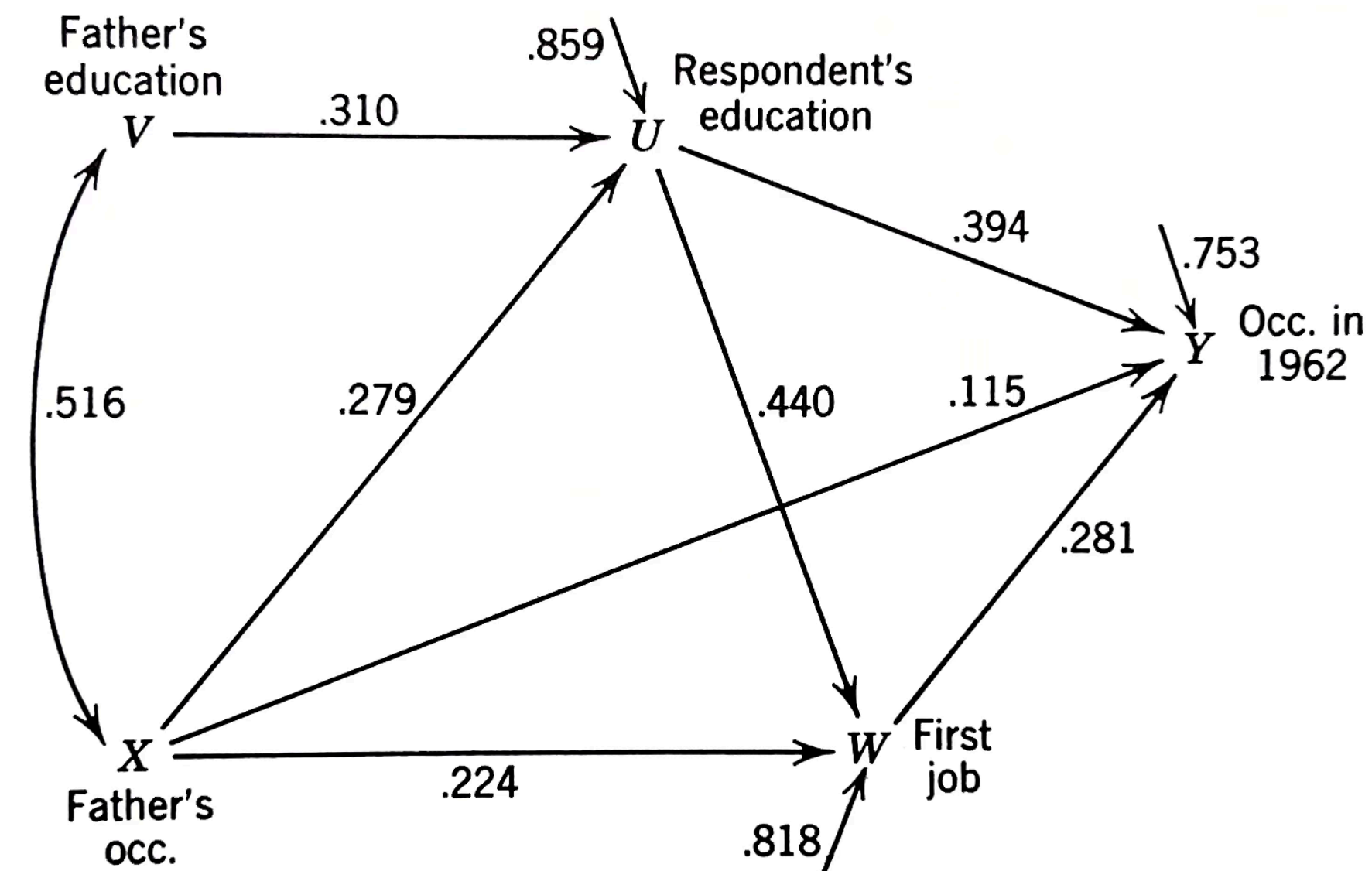


Figure 5.1. Path coefficients in basic model of the process of stratification.

Variable Identification

- Let's try for this claim:
"Improvements in people's educational attainment increases their income in the future."

Individual Work

Assignment #2a

- revise your RQ and conceptualization + your methods (briefly)
- your IV? (Mediator[s]?) DV?
- potential Confounders?
- alternative IV(s) & DV(s)? additional Mediators and/or Moderators?

Assignment #2a

- due next Wednesday, Sept. 23
- submit 2 things:
 - your variables + conceptualizations
 - master list as appendix
- This prepares for Assignment #2b, which is a group and will be graded in more detail
- *Note: you don't have to cite for everything this time*

Check Out

- Please fill in this Check Out Form before you leave
- I will be using this form to track your attendance. This form is also a place where you can share your thoughts, questions, concerns, etc. You can also share your suggestions on how I can improve our sections.
- <https://forms.gle/QKiE6zYWX35XSEZ38>

